Wedding & Event Program Management System — Database Schema

Domain 1 — Identity & Authentication

`USERS`

Stores all system accounts regardless of role.

Column	Type	Constraints	Notes
id	UUID	PK	
username	VARCHAR(80)	UNIQUE, NOT NULL	Used for login
email	VARCHAR(255)	UNIQUE, NOT NULL	
password_hash	TEXT	NULLABLE	Null when using OAuth only
auth_provider	VARCHAR(20)	DEFAULT 'local'	'local', 'google'
oauth_token	TEXT	NULLABLE	Encrypted Google OAuth2 token
is_active	BOOLEAN	DEFAULT TRUE	Soft disable account
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Purpose: Single source for all human actors — system admins, org owners, staff users.

`ROLES`

Named role definitions (Admin, Staff, Viewer, etc.).

Column	Type	Constraints
id	UUID	PK
name	VARCHAR(50)	UNIQUE, NOT NULL
description	TEXT	NULLABLE

`PERMISSIONS`

Atomic action-resource pairs.

Column	Type	Constraints	Example values
id	UUID	PK	
action	VARCHAR(30)	NOT NULL	'create', 'read', 'update', 'delete'
resource	VARCHAR(50)	NOT NULL	'guests', 'gift_records', 'reports'

Composite UNIQUE on (action, resource).

`ROLE_PERMISSIONS` ***(junction)***

Maps roles to their allowed permissions. PK = (role_id, permission_id).

Column	Type	Constraints
role_id	UUID	FK → ROLES
permission_id	UUID	FK → PERMISSIONS

`USER_ROLES` ***(junction)***

Assigns roles to users, scoped to organization context (optional).

Column	Type	Constraints
user_id	UUID	FK → USERS
role_id	UUID	FK → ROLES
assigned_by	UUID	FK → USERS
assigned_at	TIMESTAMPTZ	NOT NULL

PK = (user_id, role_id).

Domain 2 — Organizations

`ORGANIZATIONS`

Represents the application/program owner entity (the business entity owning programs).

Column	Type	Constraints	Notes
id	UUID	PK	
owner_user_id	UUID	FK → USERS	Primary admin
name	VARCHAR(200)	NOT NULL	
slug	VARCHAR(100)	UNIQUE, NOT NULL	URL-friendly identifier
telegram_bot_token	TEXT	NULLABLE	Encrypted
telegram_group_id	VARCHAR(100)	NULLABLE	For group notifications
google_drive_folder_id	TEXT	NULLABLE	Root Drive folder
google_oauth_token	TEXT	NULLABLE	Encrypted refresh token
created_at	TIMESTAMPTZ	NOT NULL	

`ORG_MEMBERS`

Staff users who belong to an organization (multi-user under one org).

Column	Type	Constraints	Notes
id	UUID	PK	
org_id	UUID	FK → ORGANIZATIONS	
user_id	UUID	FK → USERS	
role	VARCHAR(30)	NOT NULL	'admin', 'staff', 'viewer'
joined_at	TIMESTAMPTZ	NOT NULL	

UNIQUE on (org_id, user_id).

Domain 3 — Programs & Templates

`PROGRAM_TEMPLATES`

Reusable starting configurations for wedding, engagement, birthday, etc.

Column	Type	Constraints	Notes
id	UUID	PK	
name	VARCHAR(100)	NOT NULL	e.g., "Digital Wedding"
type	VARCHAR(50)	NOT NULL	'wedding', 'birthday', 'corporate'
default_fields	JSONB	NULLABLE	Template-specific field defaults
created_at	TIMESTAMPTZ	NOT NULL	

`PROGRAMS`

Core entity — one record per event/program created by an org.

Column	Type	Constraints	Notes
id	UUID	PK	
org_id	UUID	FK → ORGANIZATIONS	
template_id	UUID	FK → PROGRAM_TEMPLATES, NULLABLE	
created_by	UUID	FK → USERS	
title	VARCHAR(300)	NOT NULL	
type	VARCHAR(50)	NOT NULL	'wedding', 'birthday', etc.
slug	VARCHAR(150)	UNIQUE, NOT NULL	Used in public URL
public_url	TEXT	GENERATED	Computed from slug
qr_code_url	TEXT	NULLABLE	Generated QR image URL
description	TEXT	NULLABLE	
event_date	DATE	NOT NULL	
venue	VARCHAR(300)	NULLABLE	
status	VARCHAR(20)	DEFAULT 'draft'	'draft', 'active', 'completed'
settings	JSONB	NULLABLE	Flexible per-program config
created_at	TIMESTAMPTZ	NOT NULL	

Domain 4 — Guests

`GUESTS`

Every invited or walk-in person linked to a program.

Column	Type	Constraints	Notes
id	UUID	PK	
program_id	UUID	FK → PROGRAMS	
name	VARCHAR(200)	NOT NULL	
gender	VARCHAR(10)	CHECK ('male','female','other')	
email	VARCHAR(255)	NULLABLE	
phone	VARCHAR(30)	NULLABLE	
telegram_chat_id	VARCHAR(100)	NULLABLE	For private Telegram DM
guest_code	VARCHAR(50)	UNIQUE	Alphanumeric code on invitation
qr_token	VARCHAR(100)	UNIQUE	Token embedded in QR code
table_number	INTEGER	NULLABLE	Denormalized shortcut (see TABLE_ASSIGNMENTS)
status	VARCHAR(20)	DEFAULT 'invited'	'invited', 'confirmed', 'attended', 'no-show'
invited_at	TIMESTAMPTZ	NULLABLE	
created_at	TIMESTAMPTZ	NOT NULL	

Note: `guest_code` and `qr_token` are both unique per guest. `guest_code` is human-readable (for manual lookup); `qr_token` is machine-readable (for scanner). Walk-in guests are recorded here with `is_walk_in` flag carried by GIFT_RECORDS.

Domain 5 — Messaging

`MESSAGE_LOGS`

Audit trail of every message sent to guests (Telegram or Email).

Column	Type	Constraints	Notes
id	UUID	PK	
program_id	UUID	FK → PROGRAMS	
guest_id	UUID	FK → GUESTS, NULLABLE	Null = broadcast
channel	VARCHAR(20)	NOT NULL	'telegram', 'email'
message_type	VARCHAR(30)	NOT NULL	'invitation', 'reminder', 'thank_you', 'album_link'
status	VARCHAR(20)	NOT NULL	'pending', 'sent', 'failed', 'delivered'
content	TEXT	NOT NULL	Rendered message body
sent_at	TIMESTAMPTZ	NULLABLE	
error_msg	TEXT	NULLABLE	

Domain 6 — Seating

TABLES

Table definitions within a program.

Column	Type	Constraints
id	UUID	PK
program_id	UUID	FK → PROGRAMS
table_number	INTEGER	NOT NULL
label	VARCHAR(100)	NULLABLE
capacity	INTEGER	NULLABLE
notes	TEXT	NULLABLE

UNIQUE on (program_id, table_number).

`TABLE_ASSIGNMENTS`

Links a guest to a table (can change over time).

Column	Type	Constraints
id	UUID	PK
table_id	UUID	FK → TABLES
guest_id	UUID	FK → GUESTS
assigned_at	TIMESTAMPTZ	NOT NULL
assigned_by	UUID	FK → USERS

UNIQUE on (guest_id) to enforce one-table-per-guest at any time.

Domain 7 — Gift Recording (Handshake)

`GIFT_RECORDS`

Records each gift/money envelope received at the event.

Column	Type	Constraints	Notes
id	UUID	PK	
program_id	UUID	FK → PROGRAMS	
guest_id	UUID	FK → GUESTS, NULLABLE	Null if pure walk-in not matched
recorded_by	UUID	FK → USERS	Staff member who entered record
amount	DECIMAL(12,2)	NOT NULL	
currency	VARCHAR(10)	DEFAULT 'KHR'	
entry_method	VARCHAR(20)	NOT NULL	'qr_scan', 'manual_search', 'new_walkin'
is_walk_in	BOOLEAN	DEFAULT FALSE	Guest not in invite list
walk_in_name	VARCHAR(200)	NULLABLE	Captured when is_walk_in=TRUE
walk_in_gender	VARCHAR(10)	NULLABLE	
notes	TEXT	NULLABLE	
is_adjusted	BOOLEAN	DEFAULT FALSE	Corrected after-the-fact entry
recorded_at	TIMESTAMPTZ	NOT NULL	Can be entered post-event

entry_method distinguishes the three cases from spec:

- `qr_scan` — guest showed QR, scanner confirmed
- `manual_search` — guest found by name search (no QR)
- `new_walkin` — entirely new, not on invite list

Domain 8 — Expenses

`EXPENSES`

Program-related costs tracked for final report.

Column	Type	Constraints	Notes
id	UUID	PK	
program_id	UUID	FK → PROGRAMS	
created_by	UUID	FK → USERS	
category	VARCHAR(100)	NOT NULL	'catering', 'decoration', 'venue', etc.
description	TEXT	NOT NULL	
amount	DECIMAL(12,2)	NOT NULL	
currency	VARCHAR(10)	DEFAULT 'KHR'	
expense_date	DATE	NOT NULL	
created_at	TIMESTAMPTZ	NOT NULL	

Domain 9 — Media

`MEDIA_ALBUMS`

References to Google Drive folders where wedding photos are stored.

Column	Type	Constraints	Notes
id	UUID	PK	
program_id	UUID	FK → PROGRAMS	
title	VARCHAR(200)	NOT NULL	
drive_folder_id	TEXT	NOT NULL	Google Drive folder ID
shareable_link	TEXT	NOT NULL	Guest-accessible link
access_level	VARCHAR(20)	DEFAULT 'anyone_with_link'	
created_at	TIMESTAMPTZ	NOT NULL	

Domain 10 — Audit

`AUDIT_LOGS`

Immutable log of all create/update/delete actions across the system.

Column	Type	Constraints
id	UUID	PK
user_id	UUID	FK → USERS
action	VARCHAR(20)	'INSERT', 'UPDATE', 'DELETE'
table_name	VARCHAR(100)	Target table name
record_id	UUID	PK of affected record
old_data	JSONB	Previous state
new_data	JSONB	New state
created_at	TIMESTAMPTZ	NOT NULL

Normalization Analysis (NF)

First Normal Form (1NF) ✅

All tables satisfy 1NF:

- Every column holds atomic values. Arrays and flexible data are intentionally stored in JSONB (e.g., `settings`, `default_fields`) — these are treated as opaque documents, not multi-valued attributes.
- Each table has a single-column primary key (UUID), except pure junction tables which use composite PKs.
- No repeating groups exist anywhere.

Second Normal Form (2NF) ✅

All non-key attributes depend on the full primary key:

- Junction tables (`ROLE_PERMISSIONS`, `USER_ROLES`) contain only foreign keys or metadata that depends on the full composite key.
- `GUESTS` is fully functionally dependent on its own `id`, not partially on `program_id`.
- `GIFT_RECORDS` stores `walk_in_name` and `walk_in_gender` which are conditional on `is_walk_in=TRUE` — this is acceptable because they apply to the full record key, not a partial key.

Third Normal Form (3NF) ✅

No transitive dependencies:

- `GUESTS.program_id → org_id` does not create a transitive issue because `org_id` is not stored on GUESTS — it is derived via join to PROGRAMS.
- `GIFT_RECORDS.program_id` and `guest_id` are direct FKs, not derived from each other.
- `PROGRAMS.public_url` should be computed (GENERATED column or application-layer) from `slug` to avoid update anomalies.
- `ORG_MEMBERS.role` is a local role scoped to that org membership — this is not a transitive dependency on ROLES table because it is a simplified enum, not a FK.

Boyce-Codd Normal Form (BCNF) ✅

- Every determinant is a candidate key. No non-trivial FDs exist where the determinant is not a superkey.
- Potential BCNF concern: `GUESTS.guest_code` is UNIQUE (a candidate key), so `guest_code → all guest fields` is valid and does not violate BCNF.

Denormalization decisions (intentional)

Column	Table	Reason
`walk_in_name`, `walk_in_gender`	GIFT_RECORDS	Walk-ins may never become full GUESTS rows; embedding avoids orphan records
`table_number`	GUESTS	Read-performance shortcut; TABLE_ASSIGNMENTS is the authoritative source
`public_url`	PROGRAMS	Should be a GENERATED column from `slug` to avoid deviation

Column	Table	Reason
`currency`	GIFT_RECORDS, EXPENSES	Denormalized per record to support multi-currency programs

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Key Indexes (recommended)

```sql
-- Guest lookup by QR or code
CREATE UNIQUE INDEX idx_guests_qr_token ON GUESTS(qr_token);
CREATE UNIQUE INDEX idx_guests_guest_code ON GUESTS(guest_code);

-- Fast gift summary per program, per gender
CREATE INDEX idx_gift_program_gender ON GIFT_RECORDS(program_id, walk_in_gender);
CREATE INDEX idx_gift_guest ON GIFT_RECORDS(guest_id);

-- Message delivery status monitoring
CREATE INDEX idx_msg_status ON MESSAGE_LOGS(program_id, status, channel);

-- Audit trail lookups
CREATE INDEX idx_audit_table_record ON AUDIT_LOGS(table_name, record_id);
CREATE INDEX idx_audit_user ON AUDIT_LOGS(user_id);

-- Program listing per org
CREATE INDEX idx_programs_org ON PROGRAMS(org_id, event_date DESC);

-- Guest listing per program, gender filter
CREATE INDEX idx_guests_program_gender ON GUESTS(program_id, gender);
```

Relationship Summary

From	To	Cardinality	Notes
USERS	ORGANIZATIONS	1 : N	One user owns many orgs
ORGANIZATIONS	ORG_MEMBERS	1 : N	Many staff per org
ORGANIZATIONS	PROGRAMS	1 : N	Many events per org

From	To	Cardinality	Notes
PROGRAM_TEMPLATES	PROGRAMS	1 : N	Template used by many programs
PROGRAMS	GUESTS	1 : N	Guest list per event
PROGRAMS	TABLES	1 : N	Seating layout per event
TABLES	TABLE_ASSIGNMENTS	1 : N	Multiple guests per table
GUESTS	TABLE_ASSIGNMENTS	1 : 1	One active table per guest
PROGRAMS	GIFT_RECORDS	1 : N	All handshake records
GUESTS	GIFT_RECORDS	1 : N	One guest may give multiple gifts
PROGRAMS	MESSAGE_LOGS	1 : N	All outbound messages
GUESTS	MESSAGE_LOGS	1 : N	Messages per guest
PROGRAMS	EXPENSES	1 : N	Cost tracking per event
PROGRAMS	MEDIA_ALBUMS	1 : N	Photo albums per event
ROLES	ROLE_PERMISSIONS	1 : N	Permission grants
USERS	USER_ROLES	M : N	Role assignment via junction
USERS	AUDIT_LOGS	1 : N	All actions logged